

1. The solution of differential equation $y - x \frac{dy}{dx} = a \left(y^2 + \frac{dy}{dx} \right)$ is - $= a \left(y^2 + \frac{dy}{dx} \right)$ is -
- (a) $(x + a)(x + ay) = cy$ (b) $(x + a)(1 - ay) = cy$
 (c) $(x + a)(1 - ay) = c$ (d) None of these
2. The solution of differential equation $yy' = x \left(\frac{y^2}{x^2} + \frac{\phi(y^2/x^2)}{\phi'(y^2/x^2)} \right)$ is
- (a) $\phi(y^2/x^2) = cx^2$ (b) $x^2\phi(y^2/x^2) = c^2y^2$
 (c) $x^2\phi(y^2/x^2) = c$ (d) $\phi(y^2/x^2) = \frac{cy}{x}$
3. Solution of $y(2xy + e^x)dx = e^x dy$ is
- (a) $yx^2 + e^x = cy$ (b) $xy^2 + e^x = cx$
 (c) $xy^2 + e^{-x} = c$ (d) None of these
4. A continuously differentiable function $\phi(x)$ in $(0, \pi)$ satisfying $y' = 1 + y^2$, $y(0) = 0 = y(\pi)$ is
- (a) $\tan x$ (b) $x(x - \pi)$
 (c) $(x - \pi)(1 - e^x)$ (d) Not possible
5. The differential equation of all parabolas whose directrices are parallel to the x-axis is
- (a) $\frac{d^3y}{dx^3} = 0$ (b) $\frac{d^3x}{dy^3} = 0$
 (c) $\frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} = 0$ (d) None of these
6. The order of the differential equation whose general solution is given by $y = (C_1 + C_2)\sin(x + C_3) - C_4e^{x+C_5}$ is
- (a) 5 (b) 4 (c) 2 (d) 3
7. The solution curve of $\frac{dy}{dx} = \frac{y^2 - 2xy - x^2}{y^2 + 2xy - x^2}$, $y(-1) = 1$ is-
- (a) A parabola (b) Ellipse (c) Circle (d) Straight line
8. The equation to the curve which is such that portion of the axis of x cut off between the origin and the tangent at any point is proportional to the ordinate of that point is
- (a) $x = y(C - K \log y)$ (b) $\log x = Ky^2 + C$
 (c) $x^2 = y(C - K \log y)$ (d) None of these
9. The degree of the differential equation whose general solution is given by $y = (C_1 + C_2) \cos(x + C_3) - C_4e^{x+C_5}$ where C_1, C_2, C_3, C_4, C_5 are arbitrary constants, is -
- (a) 5 (b) 4 (c) 1 (d) 2
10. (a)

We have, $y = \left(x + \sqrt{1+x^2} \right)^n$... (i)

Let $\frac{d^2y}{dx^2} = y_2$

and $\frac{dy}{dx} = y_1$

On differentiating Equation (i), we get

$$\frac{dy}{dx} = n \left[x + \sqrt{1+x^2} \right]^{n-1} \left(1 + \frac{x}{\sqrt{x^2+1}} \right)$$

$$= \frac{n \left[x + \sqrt{1+x^2} \right]^n}{\sqrt{1+x^2}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{ny}{\sqrt{1+x^2}}$$

$$y_1^2 (1+x^2) = n^2 y^2$$

Again differentiating, we get

$$2y_1 y_2 (1+x^2) + 2x y_1^2 = 2n^2 y y_1$$

Dividing by $2y_1$

$$y_2 (1+x^2) + x y_1 = n^2 y$$

$$\Rightarrow \frac{d^2 y}{dx^2} (1+x^2) + x \frac{dy}{dx} = n^2 y$$

11. A ray of light coming from origin after reflection at the point P (x, y) of any curve becomes parallel to x-axis, the equation of the curve may be -

(a) $y^2 = x$ (b) $y^2 = 2x + 1$ (c) $y^2 = 4x$ (d) $y^2 = 4x + 1$

12. Solution of differential equation $t = 1 + (ty) \frac{dy}{dt} + \frac{(ty)^2}{2!} \left(\frac{dy}{dt} \right)^2 + \dots \infty$ is -

(a) $y = \pm \sqrt{(\log t)^2 + c}$

(b) $ty = t^{y+c}$

(c) $y = \log t + c$

(d) $y = (\log t)^2 + c$

13. The solution curve of the differential equation,

$$(x dx + y dy) \sqrt{x^2 + y^2} = (x dy - y dx) \sqrt{1 - x^2 - y^2} \text{ are -}$$

- (a) Circles of radius 1 through the origin
(b) Circles of radius 1/2 through the origin
(c) Circles not through the origin
(d) Not the circles

14. Equation of curve through (2, 2) satisfying

$$(1-x^2) \frac{dy}{dx} + xy = 5x \text{ is}$$

(a) $(x-5)^2 + 3(1-y^2) = 0$

(b) $(y-5)^2 + (1-x^2) = 3$

(c) $(y-5)^2 + 3(1-x^2) = 0$

(d) $(1-x^2) + 3(y-5)^2 = 0$

15. The solution of $\frac{x^3 dx + yx^2 dy}{\sqrt{x^2 + y^2}} = y dx - x dy$ is:

(a) $\sqrt{x^2 + y^2} = Cx$

(b) $\sqrt{x^2 + y^2} + y/x = C$

(c) $\sqrt{x^2 + y^2} + y/x^2 = C$

(d) $(x^2 + y^2)^2 + xy^2 = C$

16. Solution of the differential equation $(x + 2y^3) \frac{dy}{dx} = y$ is -

(a) $x = y^2 (c + y^2)$

(b) $x = y(c - y^2)$

(c) $x = 2y (c - y^2)$

(d) $x = y (c + y^2)$

17. The equation of the curve satisfying the equation $(xy - x^2) \frac{dy}{dx} = y^2$ and passing through the point (-1, 1), is -

- (a) $y = (\log y - 1) x$ (b) $y = (\log y + 1) x$
(c) $x = (\log x - 1) y$ (d) $x = (\log x + 1) y$

18. If m and n are the order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^5 + 4 \frac{\left(\frac{d^2y}{dx^2}\right)^3}{\left(\frac{d^3y}{dx^3}\right)} + \frac{d^3y}{dx^3} = x^2 - 1$ then :
- (a) $m = 3, n = 5$ (b) $m = 3, n = 1$
(c) $m = 3, n = 3$ (d) $m = 3, n = 2$

19. If m and n are the order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^5 + 4 \frac{\left(\frac{d^2y}{dx^2}\right)^3}{\left(\frac{d^3y}{dx^3}\right)} + \frac{d^3y}{dx^3} = x^2 - 1$, then -
- (a) $m = 3, n = 1$ (b) $m = 3, n = 3$
(c) $m = 3, n = 2$ (d) $m = 2, n = 3$

20. The differential equation which represents the family of curves $y = e^{cx}$ is
- (a) $y' = cy$ (b) $xy' - \log y = 0$
(c) $x \log y = yy'$ (d) $y \log y = xy'$